José Hiram Soltren

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Synopsis

Staff System Software Engineer, Advanced Dev & API Architecture

Proven problem-solver specializing in high-performance C++20 and Linux application development from the silicon level through the user-mode stack. Expert in designing foundational APIs, multi-threaded firmware, and highly concurrent distributed systems. Passionate about delivering ultra-reliable software and resolving complex, system-level architecture challenges.

Experience

Connectivity Engineer ForeFlight — Jan 2024 - Nov 2024

- Designed and implemented a high-performance UDP-based protocol and API for generic third-party avionics integration, successfully adopted in production by external partners (e.g., uAvionix).
- Engineered a highly optimized payload structure utilizing a novel 4-bit integer format to encode ARINC 424 flight plans, maximizing data transfer efficiency and reliability.
- Resolved a critical, FAA-escalated weather parsing outage by rapidly debugging and fixing a decade-old database query defect within one week of taking ownership, restoring safe flight operations for offshore helicopter fleets.
- Spearheaded the technical infrastructure for an on-site avionics lab, physically integrating hardware from multiple manufacturers to rapidly diagnose and deploy fixes for complex flight plan and data transfer bugs.

Technical Lead, Build and Release Roku — Jan 2023 - Sep 2023

- Architected self-service CI/CD infrastructure and implemented pre-merge CI, successfully reducing support tickets by 30% and presenting findings at IEEE ICST 2023.

Senior Software Engineer, Roku Players Roku — Jun 2020 - Dec 2022

- Architected and delivered high-performance, multi-threaded Linux applications for RokuOS firmware using C++20, ensuring ultra-reliable video decode and multimedia playback.
- Spearheaded Apple AirPlay and YouTube certification across all HDMI player platforms, designing fundamental system APIs and refactoring legacy code to satisfy strict third-party partner requirements.
- Served as Firmware Lead for the Roku Streaming Stick 4K (20M+ units); embedded with the hardware team to root-cause and resolve a critical device brownout defect by debugging and fixing an errant SoC power configuration.
- Engineered a novel, system-level debug flow—synchronizing multi-source logs, timestamps, and video frame captures alongside gdb and git—to methodically resolve complex, multi-process software defects.
- Leveraged a highly detail-oriented approach within the internal alpha program to identify and resolve critical video quality bugs across middleware layers, bridging RokuOS software and SoC vendor code.

Deep Learning Research Engineer Centaur Technology — Summer 2019

- Developed high-performance PyTorch and TensorFlow extensions for a proprietary AI inference accelerator, optimizing machine learning workloads for custom hardware.

Deep Learning Software Engineer Nervana / Intel Corp. Aug 2017 - Feb 2019

- Engineered high-performance C++ firmware for an early-generation AI training hardware accelerator, optimizing the hardware-software interface for massive-scale deep learning workloads.
- Invented and implemented a novel, highly parallel all-to-all data transfer algorithm that outperformed the industry-standard Bruck Algorithm, drastically reducing network latency across the training cluster.
- Demonstrated a strong bias for action by aggressively debugging and resolving complex, low-level hardware synchronization and memory defects to ensure reliable execution of GEMM and convolution operations.

Software Engineer Cloudera, Inc. May 2016 - Aug 2017

- Engineered distributed fault tolerance and advanced parallelism mechanisms directly within Apache Spark internals (Scala/JVM), presenting architectural findings at Spark Summit.
- Developed robust tooling for cluster analysis and automated fault detection, identifying bottlenecks and mitigating failure modes in large-scale concurrent environments.

Senior System Software Engineer NVIDIA Corporation May 2011 - Apr 2016

- Served as a key member of the Linux graphics driver team, earning a promotion to Senior Engineer through sustained technical delivery of high-performance, user-mode C/C++ code.
- Acted as Technical Lead for the VDPAU (Video Decode and Presentation API for Unix) stack. Championed and released an open-source VDPAU H.265 parser, successfully navigating a heavily closed-source corporate environment to deliver next-generation media capabilities.
- Executed deep system-level debugging within the Linux X11 driver, root-causing and resolving complex memory allocation bottlenecks to ensure ultra-reliable rendering and system stability.
- Investigated and resolved intricate concurrency and parallelism defects across OpenGL and CUDA architectures, optimizing hardware acceleration for highly threaded graphics and compute workloads.

Additional Activities

Commercial Pilot & Flight Instructor Soltren Consulting Apr 2024 - present

- Systems Mastery: Logged over 1,250 flight hours operating safety-critical aerospace systems, including 500+ hours in high-performance aircraft and extensive expertise in complex Garmin avionics suites.
- Technical Excellence: Demonstrated extreme attention to detail and a strong track record of success across numerous rigorous FAA technical exams and practical flight evaluations.
- Mentorship & Leadership: Delivered 670+ hours of advanced instruction, specializing in mentoring commercial pilots through complex aircraft transitions and instructor ratings.
- Operational Execution: Serve as a Volunteer Mission Pilot and Instructor for the Civil Air Patrol, executing high-stakes flight profiles under stringent operational and compliance guidelines.

Education

Massachusetts Institute of Technology (MIT)
Master of Engineering (M.Eng.) & Bachelor of Science (S.B.)
Electrical Engineering and Computer Science

Citizenship

United States

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